

Pure CAR states admit two-sided inner intertwining sequences

MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure

theorem

Finite-set vector-state transport supplies explicit corrections; their summable conjugation bounds and state estimates yield the required two-sided sequence.

Statement

Let A be the completed CAR algebra, the norm completion of the matrix stages $A_n = M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$, with the fixed coordinate reindexing. A state means a positive continuous complex-linear functional taking value 1 at the identity; a pure state is an extreme point of this convex state space. For a unitary $u \in A$, write $\text{Ad}(u)(a) = uau^*$. For every pair of pure states φ, ψ on A , there is a sequence $\alpha_n = \text{Ad}(u_n)$ such that, for every $a \in A$, both $(\alpha_n(a))$ and $(\alpha_n^{-1}(a))$ are norm-Cauchy, and $\varphi(\alpha_n(a)) \rightarrow \psi(a)$.

Assumptions

The input functionals are pure states on the fixed completed CAR algebra. Their GNS representations and unit cyclic vectors are constructed from these states; no identification of their Hilbert spaces, equivalence of their representations, or initial closeness of the two states is assumed.

Conclusion

The required inner intertwining sequence exists for every pure-state pair. Its inverse condition concerns the actual inverses of the produced automorphisms. This is the supplier used by the separate limit theorem.

The implementation also records paths for the local corrections, but this selected theorem only concludes the sequence property. Its proof does not presume a transporting endpoint automorphism or a general KOS principle. All choices below are made once within this construction.

Proof route

Use the finite-set vector-state transport theorem to construct unitaries l_n, r_n . Put $L_n(a) = l_n^* a l_n$, $R_n(a) = r_n^* a r_n$, and choose a dense sequence (d_j) . The corrections are chosen to give, for $j \leq n$,

$$\begin{aligned} \|Q_{n+1}(d_j) - Q_n(d_j)\| &< 2^{-n} \quad (Q = L, L^{-1}, R, R^{-1}), \\ |\varphi(L_{n+1}(R_n^{-1}(d_j))) - \psi(d_j)| &< 2^{-n}. \end{aligned}$$

The first estimates make the four sequences pointwise Cauchy. The second gives the state convergence for $\alpha_n = L_{n+1} \circ R_n^{-1} = \text{Ad}(l_{n+1}^* r_n)$. The construction below records the local theorem's hypothesis, the correction it supplies, and the inequality that enables the next application.

▼ Proof steps

1. **The local estimate used in every correction.** Put $\theta_v(a) = v^*av$. Fix an irreducible representation π , a finite set $P \subset A$, and $\varepsilon > 0$. The finite-set transport theorem gives a finite matrix-unit test set T and $\delta > 0$. If μ is a unit vector state of π and ν is a unit vector state of another representation, its hypothesis is

$$|\mu(t) - \nu(t)| < \delta \quad (t \in T).$$

Under this hypothesis, **for every** finite $F \subset A$ and $\gamma > 0$, it gives a unitary v such that

$$\begin{aligned} |\mu(\theta_v(b)) - \nu(b)| &< \gamma & (b \in F), \\ \|\theta_v(a) - a\| &< \varepsilon & (a \in P), \\ \|\theta_v^{-1}(a) - a\| &< \varepsilon & (a \in P). \end{aligned}$$

Here $\mu \circ \theta_v$ is the state of $\pi(v)\zeta$ when μ is the state of ζ . The tests T, δ are chosen before F, γ . The path supplied by the theorem is not needed for this sequence conclusion.

2. **The quantities to be constructed and the induction hypothesis.** Take the irreducible GNS representations π_φ, π_ψ with unit cyclic vectors ξ, η . Choose a dense sequence (d_j) in A , and put $D_n = \{d_0, \dots, d_n\}$ and $\varepsilon_n = 2^{-n}$. For unitaries l_n, r_n , write

$$\begin{aligned} L_n(a) &= l_n^* a l_n, & R_n(a) &= r_n^* a r_n, \\ \varphi_n &= \varphi \circ L_n, & \psi_n &= \psi \circ R_n. \end{aligned}$$

These are the vector states of $\pi_\varphi(l_n)\xi$ and $\pi_\psi(r_n)\eta$. To control both an automorphism and its inverse, use

$$P_n^\varphi = D_n \cup L_n^{-1}(D_n), \quad P_n^\psi = D_n \cup R_n^{-1}(D_n).$$

Apply the local theorem to $(\pi_\varphi, P_n^\varphi, \varepsilon_n)$, obtaining $T_n^\varphi, \delta_n^\varphi$. The induction hypothesis is the explicit inequality

$$|\varphi_n(t) - \psi_n(t)| < \delta_n^\varphi \quad (t \in T_n^\varphi).$$

It enables the local theorem to change φ_n toward ψ_n with any later finite request. Purity is used to make **both** GNS representations irreducible; they will be used in separate applications of the theorem.

3. **Initialize the induction without assuming that φ and ψ are close.** First obtain $\widehat{T}, \widehat{\delta}$ from the local theorem for $(\pi_\psi, D_0, \varepsilon_0)$. The unrestricted approximation theorem supplies l_0 with

$$|\varphi(\theta_{l_0}(t)) - \psi(t)| < \widehat{\delta} \quad (t \in \widehat{T}).$$

Now $L_0 = \theta_{l_0}$ is fixed. Obtain $T_0^\varphi, \delta_0^\varphi$ for $(\pi_\varphi, P_0^\varphi, \varepsilon_0)$. The displayed inequality enables the π_ψ theorem with $\mu = \psi$ and $\nu = \varphi_0$. Request $F = T_0^\varphi$ and $\gamma = \delta_0^\varphi$. Its output r_0 satisfies

$$|\psi(\theta_{r_0}(t)) - \varphi_0(t)| < \delta_0^\varphi \quad (t \in T_0^\varphi).$$

With $R_0 = \theta_{r_0}$, this is exactly the induction hypothesis at $n = 0$.

4. **Construct l_{n+1} and obtain the estimate that will imply state convergence.** Assume the induction hypothesis at n . First obtain T_n^ψ, δ_n^ψ from the local theorem for $(\pi_\psi, P_n^\psi, \varepsilon_n)$. Apply the already enabled π_φ theorem to $\mu = \varphi_n, \nu = \psi_n$, with

$$F = T_n^\psi \cup R_n^{-1}(D_n), \quad \gamma = \min\{\delta_n^\psi, \varepsilon_n\}.$$

Let v_n be its output, and define

$$l_{n+1} = v_n l_n, \quad L_{n+1} = L_n \circ \theta_{v_n}, \quad \varphi_{n+1} = \varphi_n \circ \theta_{v_n}.$$

The theorem gives the two estimates

$$\begin{aligned} \|\theta_{v_n}^{\pm 1}(a) - a\| &< \varepsilon_n & (a \in P_n^\varphi), \\ |\varphi_{n+1}(b) - \psi_n(b)| &< \min\{\delta_n^\psi, \varepsilon_n\} & (b \in F). \end{aligned}$$

In the first line both signs are required. Restricting the second line to T_n^ψ gives

$$|\psi_n(t) - \varphi_{n+1}(t)| < \delta_n^\psi \quad (t \in T_n^\psi),$$

which is **precisely the hypothesis for the next π_ψ application**. Restricting it instead to $R_n^{-1}(D_n)$ gives

$$\boxed{|\varphi_{n+1}(R_n^{-1}(d_j)) - \psi_n(R_n^{-1}(d_j))| < \varepsilon_n \quad (j \leq n).}$$

This second restriction, not merely closeness on the matrix units, will give $\varphi(\alpha_n(d_j)) \rightarrow \psi(d_j)$.

5. Construct r_{n+1} and verify the next induction hypothesis. Since l_{n+1} is now fixed, obtain $T_{n+1}^\varphi, \delta_{n+1}^\varphi$ for $(\pi_\varphi, P_{n+1}^\varphi, \varepsilon_{n+1})$. Apply the π_ψ theorem enabled in the preceding step, using

$$\mu = \psi_n, \quad \nu = \varphi_{n+1}, \quad F = T_{n+1}^\varphi, \quad \gamma = \delta_{n+1}^\varphi.$$

For its output w_n , set

$$r_{n+1} = w_n r_n, \quad R_{n+1} = R_n \circ \theta_{w_n}, \quad \psi_{n+1} = \psi_n \circ \theta_{w_n}.$$

The conclusions are

$$\begin{aligned} \|\theta_{w_n}^{\pm 1}(a) - a\| &< \varepsilon_n & (a \in P_n^\psi), \\ |\psi_{n+1}(t) - \varphi_{n+1}(t)| &< \delta_{n+1}^\varphi & (t \in T_{n+1}^\varphi). \end{aligned}$$

The second line is the induction hypothesis at $n+1$. Thus every correction is supplied by the finite-set transport theorem, and every required stage-test hypothesis is the preceding correction's conclusion. The test sets are always chosen before the correction that must meet them.

6. Derive all four Cauchy estimates. For $j \leq n$, protection of d_j and $L_n^{-1}(d_j)$ gives, respectively,

$$\begin{aligned} \|L_{n+1}(d_j) - L_n(d_j)\| &= \|v_n^* d_j v_n - d_j\| < \varepsilon_n, \\ \|L_{n+1}^{-1}(d_j) - L_n^{-1}(d_j)\| &= \|v_n L_n^{-1}(d_j) v_n^* - L_n^{-1}(d_j)\| < \varepsilon_n. \end{aligned}$$

The first equality uses that L_n is isometric. The second explains why $L_n^{-1}(D_n)$ was included in P_n^φ . The w_n estimates give the same two bounds for R_n and R_n^{-1} . Consequently, for any of the four sequences $Q_n = L_n, L_n^{-1}, R_n, R_n^{-1}$ and $q > p \geq j$,

$$\|Q_q(d_j) - Q_p(d_j)\| \leq \sum_{k=p}^{q-1} 2^{-k} \leq 2^{1-p}.$$

For arbitrary $a \in A$, isometry then gives

$$\|Q_q(a) - Q_p(a)\| \leq 2\|a - d_j\| + 2^{1-p} \quad (q > p \geq j).$$

First approximate a by a fixed d_j , then let $p \rightarrow \infty$. This proves the four pointwise Cauchy assertions. It does not assert norm convergence of the unitaries l_n, r_n .

7. Form the inner automorphisms and check their actual inverses. Define

$$\begin{aligned} u_n &= l_{n+1}^* r_n, \\ \alpha_n(a) &= u_n a u_n^* = L_{n+1}(R_n^{-1}(a)), \\ \alpha_n^{-1}(a) &= R_n(L_{n+1}^{-1}(a)). \end{aligned}$$

The previous step implies pointwise convergence of each of the four sequences, since A is complete. To see explicitly that the two compositions are Cauchy, use the following estimate. If F_n, G_n are pointwise Cauchy isometries, let $b = \lim_n G_n(a)$ and $c = \lim_n F_n(b)$. Then

$$\|F_n(G_n(a)) - c\| \leq \|G_n(a) - b\| + \|F_n(b) - c\| \rightarrow 0.$$

Apply this once with $(F_n, G_n) = (L_{n+1}, R_n^{-1})$ and once with $(F_n, G_n) = (R_n, L_{n+1}^{-1})$. It proves the required Cauchy conditions for $\alpha_n(a)$ and its **actual inverse** $\alpha_n^{-1}(a)$.

8. Obtain the state convergence by substitution. For $j \leq n$, the boxed estimate in the construction of l_{n+1} becomes

$$\begin{aligned} |\varphi(\alpha_n(d_j)) - \psi(d_j)| &= |\varphi(L_{n+1}(R_n^{-1}(d_j))) - \psi(R_n(R_n^{-1}(d_j)))| \\ &= |\varphi_{n+1}(R_n^{-1}(d_j)) - \psi_n(R_n^{-1}(d_j))| \\ &< 2^{-n}. \end{aligned}$$

This is why the requested set contained $R_n^{-1}(D_n)$ and why the output pairs L_{n+1} with R_n , rather than with R_{n+1} . For every $a \in A$, contractivity of the states and isometry of α_n give

$$|\varphi(\alpha_n(a)) - \psi(a)| \leq 2\|a - d_j\| + 2^{-n} \quad (n \geq j).$$

Density, followed by $n \rightarrow \infty$, yields $\varphi(\alpha_n(a)) \rightarrow \psi(a)$. Together with innerness and the two Cauchy conditions, this is exactly `HasInnerIntertwiningSequence`(φ, ψ).

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure`
- GNS vector-functional supplier with both representations irreducible —
`MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_vectorFunctional`
- Main tool: state approximation with a protected finite set —
`MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx`

- Unrestricted finite-set approximation used to initialize the recursion —
MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx
- Ordered alternating corrections — MathlibAnnex.CStarAlgebra.CAR.nonempty_transition
- Initialization without initial state closeness —
MathlibAnnex.CStarAlgebra.CAR.nonempty_initialState
- Forward output is pointwise Cauchy —
MathlibAnnex.CStarAlgebra.CAR.outputAutomorphisms_cauchy
- Actual inverse output is pointwise Cauchy —
MathlibAnnex.CStarAlgebra.CAR.outputAutomorphisms_symm_cauchy
- State control on dense tests —
MathlibAnnex.CStarAlgebra.CAR.outputAutomorphisms_state_dense
- Extension of state convergence to all elements —
MathlibAnnex.CStarAlgebra.CAR.tendsto_outputAutomorphisms_state
- Purity gives irreducibility of the GNS representation —
MathlibAnnex.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom
- The canonical GNS vector has norm one — MathlibAnnex.CStarAlgebra.norm_stateGNSVector
- The GNS vector functional is the original state —
MathlibAnnex.CStarAlgebra.inner_gnsStarAlgHom_stateGNSVector

Supporting route explanation

Lean source signature (exact)

```
theorem hasInnerIntertwiningSequence_of_pure
  (phi psi : Limit →L[ℂ] ℂ)
  (hphi : MathlibAnnex.CStarAlgebra.IsPureState Limit phi) (hpsi :
MathlibAnnex.CStarAlgebra.IsPureState Limit psi) :
  HasInnerIntertwiningSequence phi psi
```

In the source	Mathematical meaning
phi psi : Limit →L[ℂ] ℂ	Given continuous complex-linear functionals φ, ψ on the CAR completion A .
hphi : IsPureState Limit phi; hpsi : IsPureState Limit psi	Both are pure states: positive, normalized, and extreme in the real convex state space.
HasInnerIntertwiningSequence phi psi	The output is one sequence α_n of star automorphisms with all following requirements: for every $a \in A$, both $\alpha_n(a)$ and its actual inverse image $\alpha_n^{-1}(a)$ are norm-Cauchy; for every n , $\alpha_n = \text{Ad}(u_n)$ for a unitary u_n ; and $\varphi(\alpha_n(a)) \rightarrow \psi(a)$ for every a . No implementing-unitary convergence is required.

In the source	Mathematical meaning
HasInnerIntertwiningSequence	This conclusion imposes no initial closeness of the states, no supplied GNS representations and no path condition. The GNS vectors and alternating corrections described in the Proof steps are constructed inside the proof.

Further source notes: In the linked GNS proof, rho, sigma are π_φ, π_ψ , and xi, eta are their unit cyclic vectors. In the transition proof, s.move is the local theorem specialized using the displayed induction hypothesis. The proof hBclose verifies the T_n^ψ inequality, and hAclose verifies the next T_{n+1}^φ inequality. These names occur in the linked proof, not in the displayed signature.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure

Accepted content SHA-256: c6d6fce2b9893aeedf2db10812f4fc28f8c84bdbc2748e40903266d7ddd1ee28

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Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73